Customer Analytics & Exploratory Data Analysis Report

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1. Project Overview

This project focuses on customer analytics using synthetic datasets and real-world exploratory data analysis (EDA). It demonstrates modular Python architecture, automated reporting, and machine learning concepts applied to customer segmentation and spending behavior.

Key features:

- Synthetic dataset generation with missing values (~5%), outliers, and duplicates.
- Automated auditing (missing value checks, duplicate detection, statistical summaries).
- Structured EDA workflow with visualizations and correlation analysis.
- Machine learning integration (K-Means clustering for segmentation).

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2. Project Architecture

The project follows a modular structure:

- **data/** – Generated CSV files
- **reports/** – Summary reports
- **src/** – Core source code (generator.py, reporter.py, main.py)
- **notebooks/** – EDA notebook
- **generate_ppt.py** – Presentation generator
- **README.md** – Documentation

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3. Dataset Description

The dataset used for EDA contains **255 records** and **14 attributes**, including:

- **Demographics:** Customer ID, Age, Gender, City, Education, Marital Status
- **Financials:** Annual Income, Spending Score, Years Employed

- **Behavioral:** Purchase Frequency, Online Visits per Month, Returned Items, Preferred Device, Last Purchase Amount

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 255 entries, 0 to 254
Data columns (total 14 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   CustomerID            255 non-null    int64
 1   Age                   255 non-null    int64
 2   Gender                255 non-null    object
 3   City                  255 non-null    object
 4   Education             243 non-null    object
 5   MaritalStatus         255 non-null    object
 6   AnnualIncome          243 non-null    float64
 7   SpendingScore         255 non-null    int64
 8   YearsEmployed         255 non-null    int64
 9   PurchaseFrequency     255 non-null    int64
 10  OnlineVisitsPerMonth  255 non-null    int64
 11  ReturnedItems         255 non-null    int64
 12  PreferredDevice       255 non-null    object
 13  LastPurchaseAmount    255 non-null    int64
dtypes: float64(1), int64(8), object(5)
memory usage: 28.0+ KB
```

Data Quality

- Missing values: 12 in *Education*, 12 in *Annual Income*
- Duplicates: Removed during cleaning (final dataset size: 250 records)
- Numeric values imputed using median

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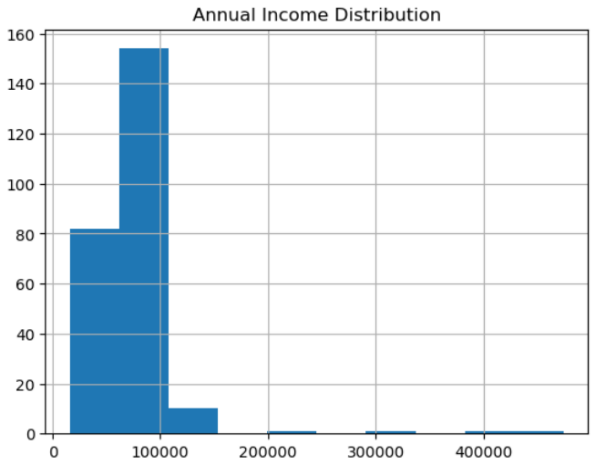
4. Exploratory Data Analysis (EDA)

4.1 Univariate Analysis

- **Annual Income:** Mean ≈ 74,500; highly skewed with outliers (max ≈ 474,000).
- **Age:** Range 21–54, median ≈ 38.
- **Spending Score:** Range 5–95, median ≈ 47.
- **Last Purchase Amount:** Mean ≈ 2,795, range 566–4,996.

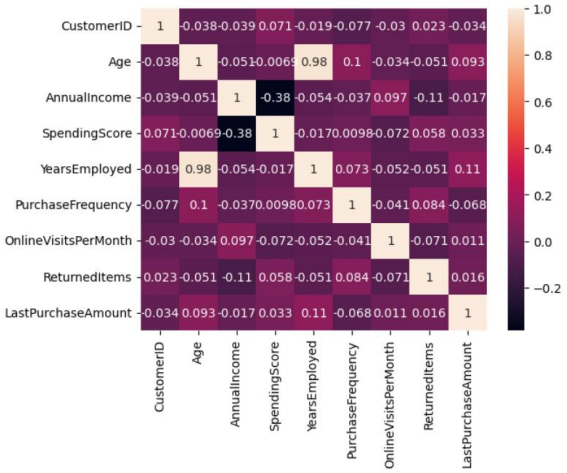


4.2 Bivariate Analysis

- **Income vs Spending Score:** Scatterplot shows weak negative correlation (≈ -0.38).
- **Age vs Years Employed:** Strong positive correlation (≈ 0.98).

4.3 Correlation Insights

- Annual Income and Spending Score are negatively correlated.
- Years Employed strongly correlates with Age.
- Purchase Frequency moderately correlates with Online Visits per Month.



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5. Machine Learning Concepts

5.1 Supervised Learning

- Uses labeled data for prediction.
- Example: Predicting customer spending score.

5.2 Unsupervised Learning

- Identifies hidden patterns in unlabeled data.
- Example: Customer segmentation using income and spending score.

5.3 Reinforcement Learning

- Learns via rewards and penalties.
- Example: Robotics, self-driving cars.

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6. Machine Learning Implementation: K-Means Clustering

Python implementation for customer segmentation:

- Segments customers into 3 clusters based on income and spending behavior.
- Useful for targeted marketing and personalized recommendations.

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7. Key Insights

- Outliers in income significantly affect distribution.
- Spending behavior shows weak correlation with income, suggesting other factors drive purchases.
- Strong relationship between age and years employed validates dataset consistency.
- K-Means clustering provides actionable customer segments for business strategy.

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8. Conclusion

The Customer Analytics Project successfully integrates:

- Modular Python architecture
- Data auditing and cleaning
- Comprehensive EDA with visual insights
- Foundational machine learning (K-Means clustering)

This framework is suitable for extending into predictive modeling, customer lifetime value analysis, and advanced segmentation for business decision-making.